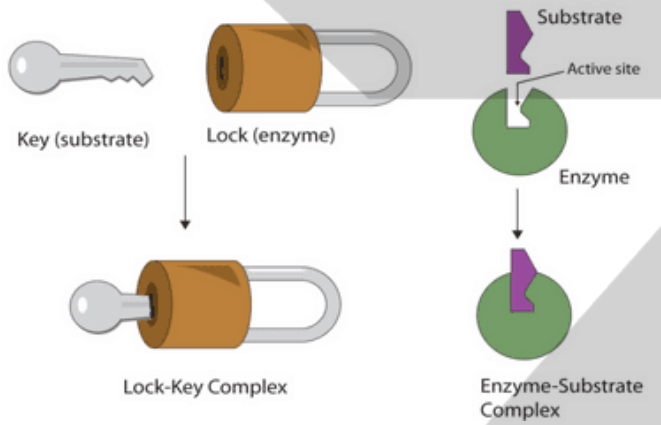
1. Enzymes

1.1. Enzymes

General Characteristics of an Enzyme

- **Catalyst**: A substance that increases the rate of reaction and is not used up itself.
- **Enzymes**: **Specific** proteins involved in all metabolic reactions, functioning as biological catalysts.
- Enzyme **lowers the activation energy** needed for a reaction to take place.
- It is essential in all living organisms for sustaining life.
- Enzymes are unchanged and can be reused

Lock and Key Model (Hypothesis)



- **Substrate:** the molecule(s) before they are made to react, **complementary** to the active site.
- **Product:** the molecule(s) that are made in a reaction

Different sequences of amino acids may lead to **varying shapes of protein molecules**, as these slight differences (temperature/pH) may be deferred in their function.

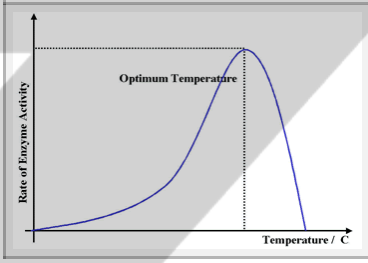
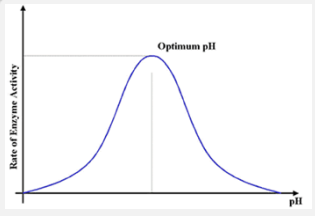
1.2. Temperature on Enzymes

- Enzymes have an optimum temperature: the temperature at which they work best. Not all animals have a body temperature of 37 °C; this temperature is a characteristic of humans, while other mammals, birds, and cold-blooded animals have different body temperatures. For example, dogs have a higher normal temperature (around (38.5 °C), birds are warmer (around (41 °C), and cold-blooded animals' temperatures vary based on their environment.
- When temperature increases, molecules **move faster, more effectively, and frequently collide**.
- Having **more kinetic energy** makes them more likely to form Enzyme-Substrate complexes.
- If the temperature is too high, enzyme molecules vibrate too vigorously; the enzyme is denatured, It loses the shape of its active site and no longer forms Enzyme-Substrate complexes.
- When the temperature is too low, there is not enough kinetic energy for the reaction, so fewer E-S complex forms

1.3. pH on Enzymes

- Enzymes are **sensitive to pH.**
- Some enzymes work best in an acid, and others in an alkaline.
- Enzymes work best at their optimum pH.
- If the pH changes, the hydrogen bond is broken, denaturing the enzyme, changing the shape of its active site. Substrate can no longer bind with the enzyme, therefore, no reaction occurs.
- Pepsin is used in acidic conditions, Amylase is used in neutral conditions, and trypsin is used in alkaline conditions.

1.4. Graphs for Changes in Enzyme Activity

Effect of Temperature	Effect of pH
	

1.6. Common enzymes and the molecules they digest

Enzyme	Substrate (what it digests)	Product	Where it is found	Where it works
Amylase	Starch	Maltose	Salivary glands, Pancreas	Mouth
Maltase	Maltose	Glucose	Small intestine	Small intestine
Lipase	Lipid (fats)	Fatty acid+ Glycerol	Pancreas, Small intestine	Small intestine
Lactase	Lactose	Glucose + Galactose	Small intestine	Small intestine
Sucrase	Sucrose	Glucose + Fructose	Small intestine	Small intestine
Protease	Protein	Amino acid	(Pepsin) Stomach	(Trypsin) Pancreas

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